

Douglass North in the crypto-economy

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Abstract.

What is the best framework for understanding the dynamics of crypto-economic systems? The dominant analytic toolset for cryptoeconomic design is game theory, mechanism design and price theory. And because blockchains are private order rules that reduce transaction costs, also New Institutional Economics (e.g. Coase, Williamson, Hart, Ostrom, *et al.*). Yet economic history is not seen as relevant. But this is wrong. A theory of history centred on institutions and organisations offers a powerful framework for studying the dynamics of crypto-economic systems. Precisely such a framework is found in the work of Douglass North, particularly *Violence and Social Orders*, a 2009 book with John Wallis and Barry Weingast. They offer a new theory of the state and of economic development centred on rent creation and the dynamics of organisations. I argue its further applicability to blockchain economies. This historical institutional perspective yields two notable conclusions: crypto-economic systems are best understood as developing economies, and the principal policy objective should be fostering their evolution towards what North *et al.* call 'open access orders'.

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1. Strange New Worlds

Our goal is to identify the most effective and comprehensive analytical framework for examining crypto-economic systems. Although the study of blockchain economics is relatively new, a best-practice consensus is emerging among economists on the use of tools from New Institutional Economics (NIE), supplemented with game theory, mechanism design (tokenomics), equilibrium approaches, price theory, and concepts from network and evolutionary economics (e.g. Abadi and Brunnermeier 2018, Berg *et al.* 2019, Catalini and Gans 2020, etc.). The emphasis on NIE primarily stems from its attention to transaction costs and private order rules – precisely what blockchains represent – which then inform our understanding of incentives, contracts, and governance mechanisms. Correspondingly, the most cited figures in crypto-economics include Nobel laureates from NIE, such as Ronald Coase, Elinor Ostrom, Oliver Williamson, and Oliver Hart, as well as others deeply concerned with coordination and rules such as FA Hayek, Thomas Schelling and James Buchanan. However, Douglass North, arguably one of the most influential NIE economists, is notably absent. This paper will show why North's historical institutional approach is particularly useful for understanding crypto-economic systems.

But why have Douglass North's significant contributions been overlooked so far? First, as an economic historian, North primarily focused on understanding economic transitions from past centuries. So it's not obvious that historical perspectives have high salience for analysing futuristic economies. Moreover North died in 2015, so he never really had a chance to observe them. Second, North's scholarship evolved over time, culminating in his 2009 collaboration with John Wallis and Barry Weingast – *Violence and Social Orders*. Yet the book remains under-recognized within the economics community. Its ambitious subtitle – *A conceptual framework for interpreting recorded human history* – suggests applicability beyond just economics, proposing a foundation for all of social science. The emphasis on 'violence' might seem irrelevant to blockchains. However, delving deeper reveals their interpretation of 'violence' as the assurance of 'security to transact with anonymity.' Recognizing this concept as a central challenge faced by societies throughout history makes North's work appear increasingly relevant and potentially the analytical cornerstone we've sought for crypto-economic systems.

Douglass North was an economic historian who pioneered the new economic history, which combined neoclassical economic theory with data driven analytics (called *cliometrics*) into an approach that, through a series of path-breaking books (*esp.* North and Thomas 1973; North 1981, 1990, 2005; North, Wallis & Weingast 2009) developed a general institutional theory of long-run economic growth and development. He won the Nobel Prize in economics for this work in 1993. For North (1991: 97) institutions are 'the humanly devised constraints that structure political, economic and social interactions'. They are both formal and informal rules and 'provide the incentive structure of the economy.' North used NIE theory and long run economic history to explain how and why institutions matter to economic performance.

North's theoretical account of historical and comparative economic performance overturned economic growth and development orthodoxy, which hitherto focused on the neoclassical economics variables of technology, capital and resources, mostly ignoring the evolutionary variables of institutions and time. North showed why this was backwards, and why the value of resources, and of investments in different types of capital or knowledge, was determined

by institutional conditions. Institutions were therefore the key thing to understand and explain. Economic historians had long understood that:

“It takes resources to define and enforce exchange agreements. ... Effective institutions raise the benefits of cooperative solutions or the costs of defection. But economic history is overwhelmingly a story of economies that failed to produce a set of economic rules of the game (with enforcement) that produced sustained economic growth. The central issue of economic history and of economic development is to account for the evolution of political and economic institutions that create an economic environment that induces in creating productivity.” (North 1991: 98)

That institutions matter is an utterly obvious point these days in economics, but only because North and colleagues worked to explain and show why.

“Economic history is about the performance of economies through time. But [neoclassical economics] focused on technological development and human capital investment but ignored the incentive structure embodied in institutions that determined the extent of societal investment in those factors. It contained two erroneous assumptions: (i) that institutions do not matter; and (ii) that time does not matter.” (North 1994: 359)

North showed why ‘institutions matter’ – in establishing the ‘rules of the game’ to shape incentive structures for cooperation and production under the division of labour – but he also explained why they were so hard to get right, why the ‘natural state’ was not modern growth, and how explaining economic change was fundamentally an explanation of the process of institutional development. The explanation is complex, involving a new theory of the state, the role of property rights and their enforcement, the deep interaction between political and economic institutions, the dynamics of organisations, and a reinterpretation of the fundamental social problem as the control of violence. This process takes place in historical time, beset with path dependencies and local contingencies.¹ North’s insights are now mainstream economics and increasingly so in development policy (e.g. see North *et al.* 2013).

I propose that blockchains and cryptoeconomic systems are a new class of objects that North and colleagues’ analytic framework might apply to. Specifically, consider them as newly discovered ‘strange new worlds’ [forgive the Star Trek reference] that have emerged very recently – sprung into existence in historical time and are, I argue, being fundamentally misunderstood in the same way that economic development was misunderstood before North. These strange new developing economies are crypto-economies in particular, e.g. built on a layer 1 blockchain, but digital economies in general, once other compute capabilities are layered onto that stack (Potts 2023). We can better understand the dynamics of these strange new blockchain economies, these crypto-economic systems, using North’s analytic framework. Moreover, a Northian framework furnishes a (much needed) new perspective on crypto-economic governance and policy. I make three broad claims that structure this paper:

- 1) blockchains compete with the state;
- 2) blockchains are developing economies;
- 3) blockchains are limited access orders.

Before we get to these claims, it will be useful to overview North’s work and agenda. Readers familiar with the NIE theory of economic history might skip the section.

¹ Of course, North was not the only economist to make this point clear, and many attribute that insight to, e.g. Hayek (1960). A similar argument about institutionally complex economic development has also been made by e.g. Rodrik (2007).

2. Douglass North's work

Douglass North was an economic historian who studied long-run economic performance. His early work was on productivity in shipping, where he showed, surprisingly, that institutional innovations mattered more than technological change (North 1968). This set him down a path exploring institutions more generally in the context of economic development. The theoretical core of what is now called 'New Institutional Economics' is the idea of transaction costs, as conceptualised by Ronald Coase to explain the existence of firms (they economise on transaction costs), and the game theoretic formulation of institutions as 'rules of the game'.

"Institutions are the humanly devised constraints that structure human interaction and they exist to reduce the ubiquitous uncertainty arising from that interaction. ... Transaction costs are the costs associated with the measurement and enforcement of agreements." North (1993: 15, 18).

The insight that firms internalise the costs of writing, bargaining, enforcement and monitoring contracts and exchange was developed by Williamson (1975, 1985). His analysis focused on hazards of opportunism in organisations due to asymmetric information and asset specificity, and studied various organisational and institutional solutions to these problems (further developed by David Teece, Oliver Hart and others). But where Williamson *et al.* used transaction costs to explain particular properties of contracts and organisations, North took transaction cost insights and applied them to entire economies, located in historical time, with particular attention to explaining the role of the state (North and Thomas 1973, North 1981). North saw organisations not so much as things to explain (*per* Coase, Williamson) but as a key part of his broader explanation, developing a somewhat Schumpeterian theory of the role of organisations as initiators of institutional change (North 1990).

"Entrepreneurs and members of organisations invest in skills and knowledge which lead to revised evaluation of opportunities, which induce alteration of the rules or gradual revision of informal constraints. The kinds of skills and knowledge perceived to have a high value payoff will reflect the incentives embodied in the institutional framework." (North 1993: 13)

"The organisations that come into existence will reflect the opportunities provided by the institutional matrix." (North 1994: 361)

The key puzzle in explaining long-run economic performance, according to North, was how property rights were established and implemented, and the problem of how governments can credibly commit to respect and enforce them. He proposed a two-step model of economic development, in which states establish property rights, and credibly commit to enforcing them in step one, bringing order and security to trade. In step two, states work to develop institutional improvements and innovations that lower transaction costs through time – e.g. that lower information costs, or enhance security and mobility of capital,² or that solve agency problems or transform uncertainty into risk – to capture gains from the growth in trade. Economic development, in this view, is not a function of natural resources (a.k.a. comparative advantage) or technological innovation *per se*, but due to the institutions that incentivise the behaviour to shape these resources into value. Sustained economic growth is

² "A capital market entails security of property rights over time and will simply not evolve where political rulers can arbitrarily seize assets or radically alter their value." (North 1991: 101)

not normal but the very thing to be explained. The long-run performance of any economy is determined by its institutions.³

North developed this argument through a series of books over many decades. In *The Rise of the Western World*, North and Thomas (1973) explored the creation and enforcement of property rights and the problem of credible commitment to do so. In *Structure and Change in Economic History* North (1981) examined how rules (in the game-theoretic sense) create property rights and the incentives for states to work to lower transaction costs in order to capture the benefits of increased security and efficiency of exchange. These works together proposed the first comprehensive property rights-based theory of social change and economic growth. In *Institutions, Institutional Change, and Economic Performance*, North (1990) addressed the question of how institutions change, recognising that mutual benefits from changing rules drives the creation of new institutions, but that high transaction costs in changing rules inhibits institutional change. John Wallis explains this as “institutions change when there are gains from doing so, but persist because of the transaction costs of changing them.” In consequence, economic dynamics are historically contingent and path dependent.

But the book most relevant to our story here is also North’s most recent, from 2009, with long-time colleagues John Wallis and Barry Weingast (hereafter NWW2009) on *Violence and Social Orders: A conceptual framework for interpreting human history*. The foundational insight of NWW2009 is that the basic social problem is overcoming violence and the key to understanding the performance of economies is the specific institutional way this problem is solved. They note that the modern theory of the state and the economy – Weber’s ‘monopoly on violence’ – does not actually explain how this works.

“Modern economics implicitly adopts the Weberian assumption that the state has a monopoly on violence and will not use it to exploit citizens, and so economics fails to understand the basic problem of development because it assumes the problem of violence away.” (NWW2009)

“The violence that matters in society is organised violence. The starting point for any theory of the state has to be how the state organises violence rather than how it uses violence. (WN 2010, 3)

All societies face the problem of violence. The institutional structure of the solution to the problem of violence is a *social order*, with each social order corresponding to a class of economies that can be built. NWW2009 propose three social orders:

1. the **forager order** – dominated hunter-gatherer society
2. the **limited access order** (or the ‘natural state’) – began 10,000 years ago
3. the **open access order** – emerged in last 300 years

NWW2009 observed that the anthropological evidence is that primitive societies in the forager order were extremely violent (Keely 1996). Civilization began with economic solutions to that problem that enabled small groups, many with specialists in violence, to mutually agree not to fight, by creating an interlocking cluster of rents, in which everyone is better off not fighting and exploiting the rents.

“Each member of the elite has exclusive or privileged control over specific resources or activities. ... Every member of the elite has an incentive to support the current dominant coalition. The rents therefore provide an incentive compatible commitment device among the elites to maintain the coalition. ... recognising and

³ Central to North’s perspective is the insight that much of economic production is shaped by the costs of exchange, i.e. transaction costs. It is only when transaction costs are zero that institutions don’t matter. Wallis and North (1986) showed that 45% of US GDP (in 1970) was devoted to the transaction sector.

supporting elite organisational forms, that is by providing the institutional framework within which elites (and no one else) can form organisations, is one of the most valuable privileges that elites possess. ... We call this blending of political and economic relationships a *natural state*." (NWW 2006)

Violence is controlled by creating *rents*, which are balanced between a dominant coalition of elites. This is, they propose, the 'natural state' of human civilizations. Most economic rents are socially constructed. But unlike costly rents in public choice theory (viz. Buchanan and Tullock 1962) the NWW2009 view implies a tradeoff between rents and stability, i.e. the cost of order is measured in acceptable or 'economically productive' rents. States administer these rents between elites in the coalition through relationships that control trade and rules that control organisations.

"Natural states do not deal with violence by consolidating control over it. Instead, utilizing the dispersion of violence in the population, they create a pattern of interlocking economic, religious, political, and social interests that provide powerful individuals with incentives not to use violence. All states are organisations of organisations. Rent-creation combines with the internal structure of organisations with the dominant coalition to limit violence in a natural state." (NWW2009)

"Organisations should be thought of as bundles of relationships, where various parties engaged in the organisation are induced to cooperate through the existence of rents created by their ongoing relationships. In order for an organisation to be self-sustaining, all members must receive rents from cooperation." (W&N 2010, 17)

By controlling access to organisations that create and exploit rents and balancing elites against each other, a government can create a mutually interlocking equilibrium with all better off owing to the rents each captures, by mutually refraining from violence. A government is a solution to the social order problem of violence, but the mechanics of the solution are a peer-to-peer system of incentives created by mutually interlocking rents, to minimise the use of violence by agents with specialist capacity to deploy violence. Violence is controlled with the invention of bureaucracy and elite privilege, not by developing an even bigger stick.

"Systematic rent-creation through limited access in a natural state is not simply a method of lining the pockets of the dominant coalition. It is the essential means of controlling violence. Rent creation and limits on both competition and access to organisations are central to the state, its institutions, and the society's performance." (NWW 2009b, 59)

"The prospect of peaceful cooperation between specialists becomes credible when both believe that the costs of fighting exceed the expected benefits. ... Gathering rents from society in turn requires elites who specialise in other activities. The individual elites in the dominant coalition depend on the regime to keep entry limited. All elites therefore have incentives to support and help maintain the coalition." (NWW 2009, 60)

"The creation of a dominant coalition is based on an interlocking set of rents, created by organisations, in which powerful individuals can credibly serve as third-parties for one another. .. The rents specialists receive from controlling their client organisations enable them to credibly commit to one another, since those rents are reduced if cooperation fails and the specialists fight." (W&N 2010, 18)

NWW2009 offers a radically new theory of social and economic dynamics in terms of two main social orders. These are

- (1) the **limited access order** (LAO), or *natural state*, which works by creating elite rents that are access controlled through organisations, and
- (2) the more modern but less common **open access order** (OAO), which is able to transcend this structure by turning rents into rights through impersonality, allowing

Schumpeterian competition through organisations. (This also gets control of the military, creating the Weberian ‘monopoly on violence’.)

A Limited Access Order is characterised by elite control of the political and economic systems to extract rents. Violence is limited because the most powerful actors maintain law and order to protect their rents. This use of rents to control violence by a state that can control organisations is what NWW2009 insist is the ‘natural state of human society. As they explicitly state: “We call this the natural state because we believe it is the natural response of human societies to the threat of endemic violence. The natural state emerged historically about 10,000 years ago and remains the dominant form of human society today.” (2006, p.15)

“Limited access orders provide a solution to violence by embedding powerful members of society in a coalition of military, political, religious, and economic elites. ... In well-developed natural states, elite privileges include control over powerful social organizations, such as the church, governments, courts, and military units. The open access order builds on the organizational achievements of the natural state but extends impersonal citizenship to an ever-growing proportion of the population. All citizens are able to form economic, political, religious, or social organizations to pursue any number of functions.” (NWW2009)

The problem, however, is this: the primary historical solution to the control of violence – maintaining law and order through elite rent creation and protection in a LAO – requires supporting economic and political monopolies, which everywhere stunts economic growth, as Schumpeter (1943) long ago made clear.⁴ A LAO requires control over organisations, including especially the ability to create them. The primary institutional cause of modern economic growth, as NWW2009 explain, is the escape from a LAO to a OAO, which is the transition to allow the impersonal creation of organisations.

“Instead of a monopoly on violence, we suggest that the governance structures of societies can be described in terms of organizational sophistication.” (NWW2009)

“Ultimately, elites moved to protect their privileges by converting them into rights through institutions that guaranteed open economic and political competition by allowing the formation of economic and political organizations at will, something the world had never seen before.” (NWW2009)

Open access orders allow any person (including legal persons) who meets some impersonal criteria to form political and economic organisations, resulting in a Schumpeterian process of creative destruction.⁵

“The systematic creation of rents can induce powerful individuals and groups to refrain from violence. When combined with the simple and powerful idea that the most valuable form of rent-creation in most societies is the ability to form organizations sanctioned and supported by the society, we were a few steps away from the implication that the structure of all social institutions is deeply conditioned by the methods used to address the problem of violence.” (NWW2009)

Open access orders thereby facilitate economic growth through the ability to create new impersonally defined, perpetually lived organisations to solve problems.⁶ Innovation, in

⁴ “In an LAO the primary concern is to reduce violence, which is a prerequisite for any economic development. When rents are eliminated, the relevant counterfactual alternative is often not a competitive market economy, but a society with disorder and violence.” (NWWW 2013, 13). “To the extent that rent creation in LAOs is the means of creating stability, then rents are a symptom of the development problem, not a cause of it.” (NWWW 2013, 5)

⁵ “The rents associated with specific organisations are what enable private ordering. The rents from specific organisations are reduced if everyone can form an organisation. This is the logic of Schumpeterian creative destruction.” (W&N 2010, 9)

⁶ “Impersonality (interaction without depending on identity) pervades open access societies, in law, markets, education, religion, politics, and the delivery of public services.” (W&N 2010, 22)

economics and in politics, requires new organisations. Impersonality is key to this story and should be distinguished from *anonymous* exchange, i.e. when the counterparty is personally unknown, but can know through membership in organisations, e.g. groups, tribe, city, organisation. Impersonal is when treated with rules. impersonal enables the creation of meaningful rights, including rights made into contracts to enable agreements to spread across space and time. A limited order society is of course capable of anonymous exchange. But a LAO is not capable of impersonal exchange. Open access orders also use democracy in different ways to LAOs. For instance, they can create public goods that do not depend on specific relations, i.e. can be impersonal. This makes it much easier to supply high-quality public goods, which is a significant source of human welfare.

“Impersonal rules about the formation of organisations meant many of the rents that could be created by privileges could be competed away. ... Just as allowing open access to organisations unleashes Schumpeterian creative economic destruction, allowing open access to political organisations unleashes creative political destruction.” (W&N 2010, 38)

“Creative political and economic destruction is the norm in open access societies. Such experimentation is precisely in the spirit of Hayek's (1952) optimistic views on the consequences of entry and competition. Adaptive efficiency entails the creation of institutions and organizations that encourage experimentation, reward successful innovation, and, equally important, eliminate failures.”

“Open access orders can only emerge when the ability to form an organisation is open to all citizens. Because the state is the organisation that organises other organisations.” (W&N 2010, 9)

The institutional transition from LAOs to OAOs *is* the process of economic development in NWW2009. The theory of economic change is this: The primal state is a ‘foraging order’ with endemic violence and consequently without economic specialisation and division of labour.⁷ Modern economies with specialisation and trade first emerged some 10,000 years ago when the problem of violence was solved with rent-creation balanced between elite coalitions. The social order solution is incentive-compatible because each now has incentive to refrain from violence in order to protect the mutually supported, but limited access, rents.

And so it went: history unfolded with interdependent economics, politics and beliefs. Different experiences and learning produced different societies and civilisations with different success solving economic problems. There was institutional variety and path dependency. The use of rent-creation to constrain violence enabled much larger societies to support larger populations, urban agglomerations and significant technological change, often militarily motivated (Turchin 2013). This is North and colleagues' history of the past 10,000 years. But they then observe that a significant series of somewhat unlikely institutional innovations occurred that had the effect of transitioning a few countries out of the natural state, making a transition from limited access orders to open access orders.

Natural states (LAOs) have limits on the type of social arrangements that can be supported. Anything that threatens rent-creation may eventually threaten the provision of order. The control of violence through rent-creation and limited access social order therefore places fundamental limits on the long-term economic growth performance of natural states. These limits became glaringly apparent over the last two centuries, when compared to open access order economies.

⁷ “The foraging order deals with violence imperfectly through the formation of groups in which personal interaction occurs among individuals who know each other well and interact repeatedly.” (NWW2009)

The transition to OAO was the centre-piece of NWW2009's long run theory of economic performance and the explanation of the modern wealth of (OAO) nations.

"it is time to acknowledge that open access societies are not just modestly improved versions of the societies that preceded them. Whereas the origins of the transition in the Western world lay in the eighteenth century and earlier, the events that transformed those societies and produced a new social order with a fundamentally different logic occurred in the mid-nineteenth century. Since then a relatively small number of societies and a small percentage of the world's population have made the transition to open access." (NWW2009)

Consider now a new idea, the main contribution of this paper – that blockchains might be another path to escape the natural state.

4. A North-Wallis-Weingast theory of blockchain economies

A blockchain economy is an institutional technology stack that establishes 'common knowledge' economic facts (e.g. of ownership, promises, rights) and facilitates the decentralised impersonal exchange of property rights composed on those facts through verified and secure transfer of electronic rights (Berg *et al.* 2019). A full-stack digital economy is a composable and 'computable' economy as an evolving complex system (Potts 2000, 2023). From this institutional technology perspective, we can recast emergent digital economies into the NWW2009 framework, starting with the idea that violence is the basic social problem to solve, and rent-based solutions to that problem held together as an incentive aligned coalition of elites is the natural state.

So first, what is a digital economy equivalent of *violence* (the social problem to solve) and the '*forager order*' (the social order that comes before the emergence of economies, history, civilization) in the natural state? I suggest the following approximation: violence (in society) = the security problem (in computers) (Miller 2006). Violence in NWW2009 means insecurity of bodily autonomy and possession, where some external party can take your property through coercion, and without voluntary agreement, through the application of force. This is like the access control problem (also the problem of bugs) in computer security (Miller and Drexler 1988). The economic consequence of violence is involuntary agency: some malicious external actor can make decisions on your behalf without your consent. Weak internet or computer security allows the same phenomena, where a malicious agent can act as if they were you. And the forager order is the pre-1993 internet, before search facilitated business models for commerce and therefore required clear property rights. Argue amongst yourselves about either of these stipulations.

Blockchains compete with the state

Key functions of the state in furnishing economic institutions – such as providing money, exchange infrastructure, asset registries, identity, courts, security (Nakamoto 2008, Abadi and Brunnermeier 2018, Szabo 2018, etc) – can be provided with a layer 1 blockchain as a private order infrastructure for building a new type of 'economy'. This is an obvious point, made by many theorists of crypto-economies (e.g. Berg et al 2019, Catalini and Gans 2020, Srinivasan 2022). These approaches all implicitly build on North's (1981) canonical economic model of the state as centred on provision and enforcement of property rights. North (1981) explains the problem thus: any organisation powerful enough to offer protection is also powerful enough to prey on citizens. It follows that credible commitment not to become a predator is the central institutional challenge of state development. (Cf. Olson's

1993 theory of the origin of the state as a ‘stationary bandit’, in which citizens legitimise the state as ‘their bandit’ to protect against external predation by ‘roving bandits’.)

“The dominant coalition creates rents for its members by limiting access to valuable resources and activities. ... The power of the state flows from its ability to organise other organisations. ... The origins of governments therefore are to serve the interests of powerful groups who are capable of sustaining formal public organisations. ... *States* are interlocking sets of organisations that can support private ordering of relationships. *Governments* are public organisations.” (W&N 2010, 4-6)

The insight that blockchains compete with the state (a public order) is made less often than its corollary: that blockchains are a *private order*, à la Ostrom on commons governance.⁸ Many researchers exploring crypto-economic governance or public goods have either been inspired by or directly used Ostrom’s analytic frameworks (e.g. Weyl et al 2022, Emmett 2022, Nabban 2022, Rennie 2023). However, explicit citation of North’s theory of economic change is less prominent, although there are a few pointed observations that blockchain consensus, smart contracts, and digital signatures are transaction cost lowering innovations in property rights that are new sources of economic progress.⁹ Frolov (2020) specifically points to North and Wallis (1986, 1988) in the context of the size of the ‘transaction sector’.

The technical argument why blockchains compete with states is this: they fulfil the role of ‘credible third-party’, which is instrumental to the mechanics of how interlocking and counterbalancing incentives work in NWW2009s natural state. Each elite is the ‘credible third-party’ of the other elites, forming a dominant coalition. Each monitors and checks the other, with each ready to move on the other if they fail, but otherwise held in dynamic balance. This is also the exact underlying mechanics of how blockchains maintain order, namely by organising for the many nodes (miners, validators, authorities, etc) to mutually watch and verify each other. Note that the dominant coalition in NWW2009 is a state or sovereign. Blockchains are sometimes described as ‘self-sovereign’ but really they are ‘peer-sovereign’. This same idea expressed in game theory format has been elsewhere described as ‘common knowledge’;¹⁰ namely that blockchains provide a public information set that is time-stamped and tamper-resistant, enabling cheap and global verification of institutional facts by individuals and between peers, and so they can be reliably and safely used as inputs to economic transactions.

But the broad argument that blockchains compete with states is a fundamentally Northian syllogism. Or to unpack, that the way to think about blockchains is as new types of economies, and the way to analyse performance of new types of economies is in terms of their institutions,¹¹ and that institutions establish and enforce property rights, and that blockchains are a new ‘institutional technology’ (Davidson *et al.* 2018) of property rights that lowers transaction costs (Novak *et al.* 2019) – that blockchains compete with states as institutional technologies is a fundamentally Northian syllogism. So, what follows?

Blockchains are developing economies

The surprising implication is this: we should think of blockchains as developing economies, in the sense proposed by NWW2009, which is that economic development is the institutional

⁸ For instance, Davidson et al 2018, Bodon et al 2019, Howell and Potgieter 2019, Rozas et al 2021, Ajwani-Ramschandani et al 2021.

⁹ See e.g. Kate Sills (on a *Kleros* podcast), Betina Warberg (in a TED talk), Eric Alston (in a series of papers, Alston 2020, Alston et al 2021), and Michael Munger (a former student of North, in his book *Tomorrow 3.0*).

¹⁰ For instance, by Silvio Micali (founder of Algorand), on a 2020 Lex Fridman podcast.

¹¹ Because transaction costs are a large fraction of the economy (Wallis and North 1986).

phase-transition from a LAO to an OAO.¹² Surprising, because the dominant narrative around blockchains (as new economic orders) is as highly technologically advanced, ‘from the future’ as it were, in a ‘stages of development’ sense. The dominant narrative is that blockchain-based digital economies are a next-generation advance on industrial economy systems of trade and finance, with administrative or operational institutional infrastructure now made of code, powered by digital computation and woven into the internet (de Filippi and Wright 2018, Werbach 2018). Indeed, the synonym for blockchain as ‘web3’, as in the third-generation of the internet, renders the idea of an evolutionary technological progression clear. So the idea that blockchain economies are instead best analysed as if they are economies from the past, or atavistic and inferior in important ways, seems jarring because the whole point is that on key operational dimensions that are massively technologically advanced. But this ‘developing economy’ view is the correct view, as I will explain, even if it appears to get the historical timeline wrong.

The way to think about it instead is as a kind of rift in time, where a new type of economy has emerged that in some dimensions is ‘from the future’, for instance cost and speed of sending a fully-settled verifiable transaction (Catalini and Gans 2020), or smart contract composability and global transaction networking. But in other dimensions is a step back in time, with the reappearance of a kind of ‘feudal order’ in governance and a wild increase in banditry. This is not unlike Clay Christenson’s *Innovator’s Dilemma*, where an innovative new product comes from outsiders only solving for one or a few key performance measures valued by niche customers, and which incumbent ‘firms’ (in this case, nation states) fail to see the opportunity.

The point to focus on here is the ways that ‘blockchain economies’ in particular (and ‘digital economies’ in a more general sense) are institutionally, even when apparently speed-running this process, still very much appropriately described as ‘developing economies’. They are much closer to primordial economies (what NWW2006 call a ‘primitive’ social order and NWW2009 call a ‘forager’ social order) than advanced modern economies. The value of bringing this analytic perspective is that it focuses our attention on the types of development problems such economies exhibit and their institutional formulation (and NWW2009 is the template for such analysis), and frees us from the intellectual trap of supposing that the problem we are trying to solve is a simple economic problem of *innovation adoption*, at the micro and macro scale, which is the policy corollary of the ‘blockchain economies are from the future’ thesis.

Blockchains are developing economies. The blockchain economic problem is, fundamentally, an economic development problem. This is why NWW2009 is important. It offers a radical new theory of economic development and therefore a new approach to solutions to this problem.

¹² The significance of this theory is that it rejects the linear model of economic development (e.g. neoclassical growth models) in which development is an accumulating consequence of investment in capital and technology, possibly moving through industrial stages, and which responds roughly monotonically to steps toward rule of law, administrative efficiency, elimination of rents and corruption, and so on (sometimes called the Washington Consensus). Instead, the NWW2009 theory of economic development finds a large set of institutional forms and capabilities consistent with various levels of economic performance, but that proper modern economic growth that separates an economy from politics only arrives with a very particular set of institutional transitions described by the open access order. Much of the world, including some nominally rich countries, have not made this transition. For NWW2009, this is as we should expect, as the ‘natural state’ of an economy is a LAO.

Blockchains are limited access orders

The properties of LAO as a social equilibrium are:

1. Control of violence through elite privileges
2. Limits on access to trade
3. Strong property rights for elites, weak for non-elites. Rule of law only for elites
4. Restrictions on entry and exit into organisations (military, economic, political, religious, educational)

NWW2009 analyse the history of economies over several thousand years. They find that these four institutions support an economic order, and that this is essentially the story of the state – i.e. a state is whatever does this, and a government is a particular form, as a public order. It can easily be seen that blockchains perform all of these functions too – *mutatis mutandis*, i.e. they provide security, define property rights, etc.

By the logic of NWW2009, a blockchain economy is a limited access order. This claim is important, because several policy implications follow. The first is structural. A LAO is a natural state, and so an institutional equilibrium. The equilibrium may be at high or low transaction level and gross value, i.e. successful or not, depending on conditions. A rich and a poor economy can both be a LAO, but both are a natural state equilibrium. A blockchain, if it is a LAO, will therefore not inevitably transition out of that state on a progressive development path; indeed, it likely will not. The natural state is dynamically stable.

The second implication is that the doorstep transition conditions (out of the natural state) offer a guide to how blockchain economies can indeed develop into modern open access economies. We should seek to design policy interventions and shape crypto regulation so as to achieve these doorstep conditions with the specific purpose of facilitating transition toward an Open Access Order regime. In other words, the LAO diagnosis provides a policy playbook to guide the proper (welfare maximising) objective of regulatory policy interventions directed at the problem of economic development of blockchain economies. This, as we review subsequently, is not how crypto-economic regulations are presently being drafted nor enacted. A NWW2009 inspired critique of much modern crypto- regulation is that it seems designed to keep blockchain economies in the natural state. This is a morally unjustifiable perspective.

Much hangs on whether a blockchain economy is indeed a natural state LAO, in the NWW2009 sense. So, how do we know?

The first major ‘tell’ of a LAO natural state is that violence is solved by creating specific rents that are controlled by elites through control of trade and through ‘limited access’ to organisations.

“All natural states control trade among states.

All natural states limit access to organisational forms.” (NWW2006, 33)

Control of trade through token access is literally what a blockchain is. It is a perfect trade controlling mechanism. A limited access order is limited precisely in the ability to create such organisations in order to access rents. If we allow that violence maps to malicious taking of (digital) property, and that security is furnished not by cryptographic barriers (i.e. encrypted

locks or firewalls, as that would be a primitive or forager ‘social order’ solution), but is provided through e.g. Nakamoto consensus (Nakamoto 2008), then the provision of security in blockchains is effectively an interlocking and mutually balancing set of organisations (nodes, miners, validators) that have rents that are valuable only to the extent that others can also capture these too. A major source of those rents that furnish security are from compute resources (mining) or placing assets at risk (validating) (Buterin 2021), which are fundamentally limited access orders due to specialised hardware or token-gated entry conditions, or regulatory privileges (due to jurisdiction), or authorised listing for delegation (Rennie 2023). Rents are also embedded in MEV and grants programs offered by, e.g. blockchain foundations seeking to develop public infrastructure. These are all elite rent-creating organisations. At the most basic level, blockchains work by operationalizing a system of rents among potentially malicious actors to stand down and choose cooperation as a higher payoff path than enacting the threats of violence that each agent is capable of. Blockchains are designed to be explicit about this mutual calculus of expectations, enforced with objectively verifiable and unswervable code. The genius of blockchain security architecture is that it effectively replicates using human governed machines the oldest most successful invention in human civilization to control violence upon which to build economies – a social order integrated with a computational machine order.

A second ‘tell’ that blockchains are LAOs is contained in their dynamics, or growth patterns, which is mostly like LAO growth, with spurts of rapid growth and often spectacular collapses, rather than the steady growth of OAO. As North (1993: 18) explains: “most of human history is not a story of economic growth but of stagnation or, at best, very modest economic growth.” While the standard narrative or comparison of blockchain projects and protocols is to entrepreneurial start-up firms (which indeed they often are), in which many and indeed most fail but a few succeed spectacularly (a distribution well described by a power laws), the dynamics of blockchain economies when we focus on consumer use and adoption (rather than token valuation) also looks a lot like LAO economies throughout history, in that the modal outcome is no growth, with movements backwards (due to bugs, attacks, rug-pulls, forks, regulatory shut-down, etc) as likely as movements forward (due to builders and developers moving in, creating valuable products, etc).

The third ‘tell’ is in relation to credible commitment and the intertwined nature of economics and politics. Blockchain governance started relatively small with the Bitcoin protocol (a small elite group of core devs), but in most protocols now, governance is a major factor in the development of a blockchain economy because it is the mechanism by which access to rents is controlled and managed. This makes protocol politics inseparable from protocol economics (Rennie et al 2022). Elections work differently in LAOs compared to OAOs, in the NWW2009 theory, due to their role in the dynamics of democracy in renegotiations of rents and in managing prospects of violence through this process.

“Elections may help stabilise a LAO by providing a visible and common knowledge means of assessing the relative strength of disparate groups, thereby reducing the likelihood that they will fight. In this sense elections increase the likelihood that adjustments in power among those with access to violence occur peacefully, making LAO more stable. ... Nevertheless, elections of this type fail to produce OAO style democracy.” (NWW 2013, 21)

“For democracy to work, elections must be embedded in an institutional environment that allows political competition to constrain politicians as well as to convey information to them. Elections in natural states typically fulfil these functions either inadequately or not at all.” (NWW2009b, 65)

Voting in blockchains can only weakly constrain elites, but is effective as a mechanism for testing and establishing legitimacy for actions done by elites (Buterin 2021, de Filippi *et al.* 2022). Furthermore, in an open access order, political dynamics occur by creating new organisations then legitimated through voting processes. But blockchain voting is more like corporate governance in managing agency risks (Davidson and Potts 2022) and in signalling agreement or rejection of rent transfers between elites (e.g. validators, or new projects), including decisions to exit the community (fork) or raid the protocol or community treasury. These so-called ‘governance attacks’ are clear evidence of the intertwined nature of economics and politics in crypto-economic systems.

5. Blockchain economic development

Blockchains are a new type of social order institution: you can build economies on and with them. The logic of this claim is that blockchains are essentially a new type of computer (a distributed public computer) that enables secure trustless asynchronous execution of digital transactions and agreements between many agents. Over the past 10,000 or so years a great variety of real world economies have evolved institutional stacks that include states, markets and organisations to facilitate the ‘distributed computing’ of transactions. But blockchains can also perform these institutional and administrative functions, in a limited way (they need to be digitally represented), but on some margins at vastly lower cost or with superior performance (Catalini and Gans 2020). Blockchains are a new type of institutional technology for building a new type of digital economy (e.g. Davidson *et al.* 2018, Berg *et al.* 2019). But what type of economy? This is our second point, which is that if we follow NWW2009, then blockchains are ‘natural state’ developing economies, and specifically limited access orders because they share many properties with LAOs. Given that, the key economic question is how do blockchain economies transform into modern open access order economies of the type long envisaged by, e.g. the cypherpunks (e.g. Mark Miller, Nick Szabo, Timothy May, et al, see Nabben 2023) and more recently by, e.g. visionary web3 founders such as Vitalek Buterin (viz. Zuzalu), Zooko Wilcox, or Eric Voorhees?

According to the NWW2009 framework, we should not expect this just to happen. Natural states are equilibriums. No endogenous forces transition them to OAOs.

"The relevant dynamics of social change for developing countries, however, lie in the logic of the natural state, not in the logic of open access orders. ... Even when societies perceive the benefits of open access, natural state coalition members perceive that simply adopting open access institutions would not only destroy their rents but would also fail to produce the desired outcome of sustained economic development." (NWW2009)

The NWW2009 framework does however provide a historically validated theory of the ‘doorstep conditions’ for transition from limited access orders to open access orders. These are:

1. Rule of law for elites extends to impersonal rules
2. Perpetual forms of organisation must be created
3. Political control of the military

What would it mean for these conditions to manifest in a digital economy? Start with the third condition. Blockchains don't have militaries, but ‘political control of’ means that the military is properly under civilian and therefore democratic control. Civilian democratic control of a blockchain economy translates to a properly decentralised public (not private) layer 1

protocol, uncompromised by powerful blocs of token-holders, nor under legislative or regulatory control of any particular government agencies or factions. It cannot be attacked from within by specialists wielding malicious capabilities.

On the first condition – impersonality, and the transformation of rents into rights that therefore extend beyond elites – public blockchains inherently have this property, in that access control is through objects not lists. However, institutional integration of blockchain economies into real world economies has in very many circumstances forced permissioning access control systems onto blockchain economies. This is usually publicly justified in terms of investor protection or other such public welfare and safety concerns, but it also has the effect of seeking to integrate blockchain economies into elite coalitions of rents. This is the natural state defending itself, variously seeking to either integrate a new coalition member, or to block it from ever joining the elite rent-production coalition, rendering it an outlaw organisation with shadowy institutional legitimacy in tax havens (e.g. Bermuda, Cayman Islands) or in partnerships with fringe economies (e.g. El Salvador, Marshall Islands).

But we can also observe blockchain economies working to transition to OAO in efforts to supply public goods, often through the layer-1 foundations.

“Open-access orders can deliver policies to citizens on an impersonal basis. This allows such orders to provide a wide range of public goods and large scale social insurance programs of the type that are missing in natural states. ... Impersonal delivery of public goods and services prevents political officials from threatening to withhold goods as a means to manipulate citizens” (NWW2009, 65)

Impersonal policies in OAO support a rich civil society by enabling all types of organisations and also enabling public goods with rule-governed access, such as social insurance for all citizens. These types of public goods strongly complement markets by lowering individual risk to market participation, or what McCloskey (2016) calls ‘the bourgeois deal’. The work in Regenerative Finance (ReFi) for instance to create generalised exchange platforms, universal access to saving technology (e.g. stablecoins), new social welfare products such as crypto-based universal basic income, among other new types of public goods (e.g. Buterin et al 2019), indicates the current effort to transition to an open access order.

“Understanding human development requires understanding how institutions shape the kind of organisations available for sustainable human cooperation. Developed societies are filled with a rich variety of complicated and sophisticated organisations capable of producing goods and services, carrying out research and development, and coordinating individual behaviour on a scale never before seen in human history.” (NWW2006: 1-2)

But it is the second condition – perpetual forms of organisation – that is the most interesting doorstep condition here, and the true source of the transition to a digital OAO economy.

“Our framework builds on three insights about organizations:

1. The structure, extent, and number of organizations in any society are intimately tied to the way that society controls violence.
2. The social technology of structuring organizations depends on personality and the identity of the individuals within the organization. The creation of impersonally defined, perpetually lived organizations whose identity is independent of the identity of their members is difficult to accomplish. But where it occurs, it fundamentally changes the possibility for relationships among individuals.
3. The existence of organizations with impersonal identities, in both the public and private sphere, is a necessary condition for the existence of impersonal relationships in the larger society.” (NWW2009)

“Competitive economic and political societies are impossible without open access to organisational forms.” (NWW2006, 73)

"Instead, the transition provides citizens with new tools, fewer restrictions, and greater scope for impersonal relations, all of which dramatically increase the gains from specialization and exchange while reducing the risk of expropriation. These changes in turn foster the growth of new organizations to exploit new opportunities. The explosion in organizations is thus a direct consequence of the transition." (NWW2009)

The relevant substitution from original-NWW2009 to digital-NWW2009 is that perpetual forms of *organisations* become perpetual forms of *economies*. The fundamental limitation to overcome is the rights-governed ability to spin up whole economies that are perpetual and impersonal. This requires a full stack of institutions to support and compose them, at the core of which are money and property rights.

An OAO digital economy is not an explosion of new organisations, *but an explosion of new economies*. It is open access to create economies that is the meaning of modern economic development in the Douglass North inspired framework proposed here.

6. Conclusion

"The seeming independence of economic and political systems in open access societies has deceived modern social science. In natural states, all big economic organizations are necessarily also political ones. In open access orders, big economic organizations typically concentrate far more on markets and are only tangentially involved in politics." (NWW2009)

One of the surprising lessons of recent years in blockchain is the growing realisation that blockchain protocols and communities need to get political and need to engage and fight back against political attacks. This was a shock to many because of the prevailing cryptopunk or crypto-utopian sentiment of exit, that what was being built was a set of institutional technologies and social tools to build alternative money and financial systems, alternative economic infrastructure. It was implicit in this conceit that these were post industrial digital economies from the future. But the argument I have made here is that blockchain economies are developing economies, and specifically conform to the NWW2009 model of a limited access order, which is the natural state. They needed to get political because they are born into a natural state in which economics and politics are deeply intertwined in coalitions of rent-production, and blockchains are coming in threatening to disrupt those rents. This is plainly visible in the core money and finance uses of blockchain, which have provoked the most strident response from the elite coalitions.

The NWW2009 framework provides a way to understand the nature of blockchain economic development and growth as that of a prospective transition to an open access order. Some of this is starting to occur, but the framework also explains the broader institutional dynamics we are observing in the way the existing elite rent producing coalitions are reacting to this rent disrupting technology. This also predicts that the institutional development pathways of blockchain economies might be very different in current LAOs (e.g. China, El Salvador, Russia, Vietnam, etc) compared to open access order countries (such as the US, Singapore and Australia, and also the various tax havens). But the central insight to draw is that the transition is not centred on growing openness to create organisations, but rather a desire to create whole economies. This is a strange point to make in the NWW2009 context of real world economies, where the institutional stack of money, law, property rights administration and so on is both a natural monopoly, but often a legitimately claimed monopoly, of the government. But digital blockchain economies can compete and innovate on precisely these services that compose the institutional stack of an economy. The crucial question is whether citizens have the right to build and deploy such economies. The future of digital open access

orders will therefore be one in which we observe not ‘an explosion of organisations’ but rather an ‘explosion of economies’.

Finally, the NWW2009 model of blockchain economies suggests a policy approach consistent with an open access order model. First, it should be clear that the current policy approach to blockchain economies in many countries, and by many regulatory agencies and legislative assemblies, is manifestly framed in LAO rent protection. That is logical in the natural state, but it should not be the policy framework in an open access order. Rather, the OAO goal should be to support and facilitate the transition of the developing digital economies toward an open access order regime. In other words, while ‘open access orders’ as developed since the mid-19th century in some countries are one path out of the ‘natural state’, blockchains are another. Regulation needs to focus on the doorstep conditions and support efforts to transition out of them, which is to say, that they should be treated like any other developing economy. A most-favoured-nation approach to blockchain economies, or just high-level recognition that these are developing economies, would be a good start.

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